Duration: 150 mins

Pre-Requisites

Learners are expected to have:

- Clear understanding of GenAI system architecture.
- Familiarity with prompting hierarchies and tool integration.
- Awareness of hallucination, prompt injection, jailbreaks, and evaluation metrics (BLEU, ROUGE, semantic similarity, LLM-as-judge).

Session Summary

This 150-minute live session will focus on strategic and operational decision-making in GenAI systems. Learners will first evaluate proprietary versus open model strategies before analysing specific model ecosystems. The session will then examine hosting architectures, deployment trade-offs, and operational feasibility.

Evaluation will be treated as a system pipeline stage rather than as a standalone metric discussion. Robustness and red-teaming will be positioned as architectural responsibilities. The session will conclude with a capstone case exercise in which learners will design an end-to-end GenAI system.

By the end of the session, learners will be able to justify model, deployment, evaluation, and robustness decisions in a structured and systematic manner.

Learning Objectives / Outcomes

By the end of this session, learners will be able to:

- Justify strategic selection between proprietary and open-weight LLM classes.
- Evaluate major model ecosystems based on workload requirements.
- Select appropriate hosting and deployment architectures.
- Design evaluation pipelines aligned to use-case objectives.

- Integrate robustness and guardrails into system architecture.
- Propose an end-to-end GenAI system design with justified trade-offs.

Brief Agenda

Learning Objective	Comments	Duration	Mode
Decide model class strategically	Proprietary vs open-weight trade-offs	30 min	Conceptual + Discussion
Choose specific model families	Ecosystem comparison and workload fit	25 min	Conceptual
Evaluate hosting feasibility	Deployment models and infrastructure trade-offs	30 min	Conceptual + Guided Analysis
Design evaluation pipelines	Evaluation as system stage	20 min	Conceptual
Integrate robustness mechanisms	Architectural guardrails and monitoring	20 min	Conceptual
Apply end-to-end reasoning	Capstone enterprise system design exercise	25 min	Guided Analysis + Discussion

Conclusion

By the end of this session, learners will:

- Understand how to select model classes strategically.
- Choose specific models based on workload and ecosystem maturity.
- Evaluate deployment feasibility across API, cloud, and self-hosted options.
- Embed evaluation and robustness into system architecture.

- Demonstrate end-to-end GenAI system design reasoning.

Learners are now equipped with architectural, strategic, and operational thinking required for GenAI system design.

Additional Reading Links

- OpenAI API & Platform Documentation
- Anthropic Claude Documentation
- Gemini 3: Google DeepMind
- Meta Llama Documentation
- HuggingFace Inference & Deployment Documentation
- Azure OpenAI Service Documentation